

A Norm-Closure Obstruction for Inner Jordan Derivations on $K(H)$

Run `fa_banach_001`, lane 14

2026-06-29

Status. `counterexample_likely_valid`.

1 Source Problem

The source paper is Jamjoom–Peralta–Siddiqui, *Jordan weak amenability and orthogonal forms on JB^* -algebras*, arXiv:1411.2712 [1]. Remark 3.16 raises the problem of determining whether, for a JB^* -algebra $\mathcal{J}$, the inner Jordan derivations on $\mathcal{J}$ are norm-dense in the set of all Jordan derivations on $\mathcal{J}$. The immediately preceding Remark 3.15 recalls that for $\mathcal{K}(H)$, where H is infinite dimensional, there are derivations into $\mathcal{K}(H)^*$ which are not inner in the Jordan sense.

Remark 3.15. Let A be a C^* -algebra, since the anti-symmetric orthogonal forms on A and the Lie Jordan derivations from A into A^* are mutually determined, we can deduce, via Goldstein's theorem (cf. Theorem 1.1), that every Lie Jordan derivation $D : A \rightarrow A^*$ is an inner derivation, i.e., a derivation given by a functional $\psi \in A^*$, that is, $D(a) = \text{adj}_\psi(a) = \psi a - a\psi$ ($a \in A$). We shall see that a finite number of functionals in the dual of a JB^* -algebra $\mathcal{J}$ and a finite collection of elements in $\mathcal{J}$, i.e., the inner Jordan derivations are not enough to determine the Lie Jordan derivations from $\mathcal{J}$ into $\mathcal{J}^*$ nor the anti-symmetric orthogonal forms on $\mathcal{J}$. Indeed, as we have commented before, there exist examples of JB^* -algebras which are not Jordan weakly amenable, that is the case of $L(H)$ and $K(H)$ when H is an infinite dimensional complex Hilbert space (cf. [25, Lemmas 4.1 and 4.3]). Actually, let $B = K(H)$ denote the ideal of all compact operators on H , and let ψ be an element in B^* whose trace is not zero. The proof of [25, Lemmas 4.1] shows that the derivation $D = \text{adj}_\psi : B \rightarrow B^*$, $a \mapsto \psi a - a\psi$ is not inner in the Jordan sense. Therefore the anti-symmetric form $V(a, b) = D(a)(b) = (\psi a - a\psi)(b) = \psi[a, b]$ cannot be represented in the form given in (3.9). A similar example holds for $B = B(H)$ (cf. [25, Lemma 4.3]).

Remark 3.16. We have already shown the existence of JBW^* -algebras which are not Jordan weakly amenable (cf. [25, Lemmas 4.1 and 4.3]). Thus, the problem of determining whether in a JB^* -algebra $\mathcal{J}$, the inner Jordan derivations on $\mathcal{J}$ are norm-dense in the set of all Jordan derivations on $\mathcal{J}$, takes on a new importance. When the problem has an affirmative answer for a JB^* -algebra $\mathcal{J}$, Theorem 3.13 allows us to approximate anti-symmetric orthogonal forms on $\mathcal{J}$ by a finite collection of functionals in $\mathcal{J}^*$ and a finite number of elements in $\mathcal{J}$. Related to this problem, we remark that Pluta and Russo recently proved that the set of inner triple derivations from a von Neumann algebra M into its predual is norm dense in the real vector space of all triple derivations, then M must be finite, and the reciprocal statement holds if M acts on a separable Hilbert space, or is a factor [34, Theorem 1]. It would be interesting to explore the connections between normal orthogonal forms and normal Jordan weak amenability or norm approximation by normal inner derivations.

2 Proof Intuition

For $\mathcal{A} = \mathcal{K}(\ell_2)$, every trace-class operator T implements an associative dual derivation

$$\delta_T(a)(b) = \text{Tr}(T(ab - ba)).$$

This is automatically a Jordan derivation when $\mathcal{A}$ is viewed with its Jordan product. A finite inner Jordan derivation has a more special form: after identifying $\mathcal{A}^*$ with the trace class, its implementer is a finite sum of trace-class commutators, hence has trace zero.

The point is that the trace-zero obstruction survives norm closure in the derivation norm. If δ_T is small, then $\text{Tr}(T)$ is small. To see this, move a large finite-dimensional block of ℓ_2 to a disjoint far-out block by a compact partial isometry. Evaluating δ_T on that partial isometry and its adjoint compares the trace of T on the two blocks; the first block captures $\text{Tr}(T)$ and the far-out block has negligible trace.

3 Counterexample

Theorem 1. *Let $H = \ell_2(\mathbb{N})$ and let $\mathcal{A} = \mathcal{K}(H)$, regarded as a JB^* -algebra with Jordan product $a \circ b = (ab + ba)/2$. Then the inner Jordan derivations from $\mathcal{A}$ into $\mathcal{A}^*$ are not norm-dense in the set of all Jordan derivations from $\mathcal{A}$ into $\mathcal{A}^*$.*

Proof. Identify $\mathcal{A}^*$ isometrically with the trace class $S_1(H)$ by $T \mapsto \psi_T$, where $\psi_T(x) = \text{Tr}(Tx)$ for $x \in \mathcal{K}(H)$. For $T \in S_1(H)$ define

$$\delta_T(a)(b) = \text{Tr}(T(ab - ba)) \quad (a, b \in \mathcal{A}).$$

Equivalently, $\delta_T(a)$ is represented by the trace-class operator $Ta - aT$. Hence

$$\|\delta_T\| = \sup_{\|a\| \leq 1} \|Ta - aT\|_1.$$

Since δ_T is an associative derivation from the C^* -algebra $\mathcal{A}$ into the dual bimodule $\mathcal{A}^*$, it is a Jordan derivation for the Jordan product:

$$\delta_T(a \circ b) = \delta_T(a) \circ b + a \circ \delta_T(b).$$

We first record the form of the inner Jordan derivations. Let $S \in S_1(H)$, let $c \in \mathcal{K}(H)$, and let $\phi = \psi_S$. The elementary inner Jordan derivation determined by ϕ and c is

$$D_{\phi,c} = [L(\phi), L(c)] : \mathcal{A} \rightarrow \mathcal{A}^*.$$

For $a, b \in \mathcal{A}$, the computation in the source paper gives

$$D_{\phi,c}(a)(b) = \phi(b \circ (c \circ a) - (c \circ b) \circ a).$$

Using $x \circ y = (xy + yx)/2$ in the associative algebra $\mathcal{K}(H)$,

$$b \circ (c \circ a) - (c \circ b) \circ a = \frac{1}{4}[c, [a, b]].$$

Therefore

$$D_{\phi,c}(a)(b) = \frac{1}{4} \text{Tr}(S[c, [a, b]]) = \frac{1}{4} \text{Tr}((Sc - cS)[a, b]) = \delta_{(Sc - cS)/4}(a)(b),$$

where we used cyclicity of the trace for products of trace-class and bounded operators. The implementer $(Sc - cS)/4$ has trace zero. Hence every finite sum of inner Jordan derivations is of the form δ_R with $\text{Tr}(R) = 0$.

Next we prove that trace is continuous with respect to the derivation norm on these implemented derivations. Fix $T \in S_1(H)$ and an orthonormal basis $(e_k)_{k \geq 1}$ of H . Let p_n be the projection onto $\text{span}\{e_1, \dots, e_n\}$. Then $\text{Tr}(Tp_n) \rightarrow \text{Tr}(T)$. Since T is trace class, after passing far enough into the basis we can choose a rank- n projection q_n orthogonal to p_n such that $\text{Tr}(Tq_n) \rightarrow 0$. Let v_n be a partial isometry with $v_n^*v_n = p_n$ and $v_nv_n^* = q_n$. Then $v_n \in \mathcal{K}(H)$ and $\|v_n\| = \|v_n^*\| = 1$. Thus

$$\delta_T(v_n)(v_n^*) = \text{Tr}(T(v_nv_n^* - v_n^*v_n)) = \text{Tr}(T(q_n - p_n)) \longrightarrow -\text{Tr}(T).$$

Consequently

$$|\text{Tr}(T)| \leq \|\delta_T\|.$$

Let P be a rank-one projection. Then $\text{Tr}(P) = 1$, so δ_P is a Jordan derivation with nonzero trace implementer. If D is a finite inner Jordan derivation, then $D = \delta_R$ for some $R \in S_1(H)$ with $\text{Tr}(R) = 0$, and the previous estimate gives

$$\|\delta_P - D\| = \|\delta_{P-R}\| \geq |\text{Tr}(P - R)| = 1.$$

The same lower bound holds for every norm limit of finite inner Jordan derivations. Therefore δ_P is not in the norm closure of the inner Jordan derivations from $\mathcal{A}$ into $\mathcal{A}^*$. $\square$

4 Verification Notes

The proof is analytic and uses no computation. The only source-specific calculation is the standard formula for a finite inner Jordan derivation, already displayed in Remark 3.15 of the source paper; the packet expands it in the special associative algebra $\mathcal{K}(H)$ and observes that its trace-class implementer has trace zero.

Novelty checks were bounded as follows. The run indexes were searched for 1411.2712, **Jordan weak amenability**, **inner Jordan derivations**, **norm dense**, **norm-dense**, **K(H)**, and **trace**. Later local papers citing [1] were checked around Jordan and triple derivation approximation statements. Web searches on 2026-06-29 for exact and close norm-density variants found no separate exact resolution.

5 Limitations and Review Recommendation

This counterexample answers the general norm-density problem negatively by one concrete JB^* -algebra, $\mathcal{K}(\ell_2)$. It does not classify which JB^* -algebras have norm-dense inner Jordan derivations, and it does not address the separate normal/predual approximation questions for JBW^* -algebras mentioned at the end of Remark 3.16.

Human review should focus on the trace-continuity estimate $|\text{Tr}(T)| \leq \|\delta_T\|$ and on the identification of finite inner Jordan derivations with trace-zero implementers.

References

- [1] F. B. Jamjoom, A. M. Peralta, and A. A. Siddiqui, *Jordan weak amenability and orthogonal forms on JB^* -algebras*, arXiv:1411.2712, 2014.